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25 July 1961
SUPERSEDING
RIAPD-603A
7 October 1959

MILITARY SPECIFICATION

HEATING AND TIE-DOWN UNIT, TRUCK-MOUNTED 762MM ROCKET, M78A1

1. SCOPE

1.1 This specification covers one type of truck-mounted heating and tie-down unit designed for heating and securing the 762MM Rocket and crated auxiliary equipment during transit.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-781	-	Strapping, Flat; Steel
CCC-C-429	-	Cloth, Cotton, Osnaburg
PPP-B-621	-	Boxes, Wood, Nailed and Lock Corner
PPP-T-60	-	Tape; Pressure Sensitive Adhesive, Water-proof - For Packaging and Sealing

FSC 1055

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Military

MIL-P-116 - Preservation, Methods of
MIL-B-121 - Barrier Material Greaseproofed, Flexible (Waterproofed)
MIL-P-130 - Paper, Wrapping, Laminated and Creped
MIL-W-6858 - Welding; Aluminum, Magnesium, Non-Hardening Steel or Alloys and Titanium, Spot, Seam and Stitch
MIL-S-6872 - Soldering Process; General Specification for
MIL-E-10062 - Engines, Spare or Installed (Other than Aircraft), Preparation for Shipment and Storage of
MIL-G-10924 - Grease, Automotive and Artillery
MIL-B-12121 - Plastic, Coating Compound, Strippable (Sprayable)
MIL-P-12841 - On-Vehicle Material (OVE) for Military Vehicles, Packaging of
MIL-B-13239 - Barrier Material, Waterproofed, Flexible all Temperature
MIL-P-14232 - Parts Equipment and Tools for Ordnance Material, Packaging of
MIL-W-14539 - Weapons Parts for Anti-Aircraft, Artillery, Rocket Launchers and Small Arms, Packaging of
MIL-C-16173 - Corrosion Preventive Compound, Solvent Cut-Back, Cold-Application
MIL-I-45607 - Inspection Equipment, Supply and Maintenance For Ordnance

STANDARDS

Military

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U. S. Military Property
MIL-STD-171 - Finishing of Metal and Wood Surfaces
MIL-STD-186 - Systems for Painting and Finishing Rocket and Guided Missile Material
MIL-STD-281 - Automobiles, Trucks, Truck Tractors and Trailers, Preservation of

DRAWINGS

Ordnance Corps

D8404498 - Generator Set, Gasoline Engine, 3.5 KW, M25
F8419330 - Kit, 762MM Rocket, M78A1
IEL8419330 - Index of Inspection Equipment Lists

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PUBLICATIONS

Ordnance Corps

LSQAP- 8419330 - Index of Supplementary Quality Assurance Provisions

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Association of American Railroads - Pamphlet Number MD-7, Rules Governing the loading of Department of Defense Material on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, Freight and Container Section, 59 East Van Buren Street, Chicago 5, Illinois.)

3. REQUIREMENTS

3.1 Materials.- Materials shall conform to the material specifications referenced on the drawings.

3.2 Dimensions.- Dimensions of parts and assemblies shall conform to the respective drawings of Drawing F8419330.

3.3 Performance and product characteristics.

3.3.1 Locking devices.- All movable parts of locking devices and fastenings provided for securing the rocket or crates, and for locking the various components of the rocket in position, shall function without interference and be positive in action.

3.3.2 Electrical system.- All wiring and components of the electrical system shall be properly installed, and connected correctly and securely, to assure continuous operation and proper functioning of the various receptacles and components. Each outlet receptacle of the two receptacle box assemblies (Part Number 8419360) shall have an output of 110 to 115 volts alternating current (AC).

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3.3.3 Saddle and base assemblies.- The saddle assemblies shall function freely in the slideways of their mating base assemblies. Metal shims or washers may be used, if necessary, when bolting the base assemblies to the truck bed to prevent warpage due to possible unevenness of the floor of the truck bed.

3.3.4 Generator set, gasoline engine.- The component generator set shall conform to Drawing D8404498, Generator Set, Gasoline Engine, 3.5 KW, M25.

3.3.5 Release assembly.- The release assembly (Part Number 8417254) shall be capable of exerting a pull of 26.6 pounds, minimum, at the remote control end of the actuating cable.

3.4 Government furnished equipment.- Unless otherwise specified, all Government furnished equipment intended as components of the heating and tie-down unit, or intended as a carrier for the heating and tie-down unit, shall be modified only to the extent required by applicable drawings. The contractor shall complete all mechanical and electrical adjustments necessary to provide proper functioning of the heating and tie-down unit. Malfunctioning of the unit, caused by improper modification, assembly or adjustment of the Government furnished equipment, shall be the contractor's responsibility. (See 6.1.)

3.5 Painting.- Unless otherwise specified, the heating and tie-down unit shall be painted in accordance with Standard MIL-STD-186. Color shall be specified by the procuring activity (see 6.2.2.).

3.6 Marking.- Marking for identification of parts and assemblies shall conform to the applicable drawings and Standard MIL-STD-130.

3.7 Welding.- Welding shall conform to Class B of Specification MIL-W-6858. Actual production of weldments shall not proceed prior to Government approval of the contractor's welding procedure and test specimens.

3.8 Soldering.- Soldering of component parts, including receptacles and leads, shall be in accordance with Specification MIL-S-6872.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to preseribed requirements.

4.2 Inspection provisions.

4.2.1 Materials and processes.- Materials and processes shall be inspected and tested as follows:

4.2.1.1 Materials.- Materials shall be tested in accordance with the material specification referenced on the applicable drawings.

4.2.1.2 Welding.- Welding shall be inspected in accordance with the applicable welding specification. The contractor shall obtain Government approval of welding procedures and test specimens prior to production welding.

4.2.1.3 Painting.- Painting, including cleaning and preliminary coatings and finishes, shall be inspected and tested in accordance with Standard MIL-STD-171.

4.2.1.4 Soldering.- Soldering shall be inspected in accordance with the applicable soldering specification.

4.2.2 Certification.- The contractor shall furnish the Government inspector certification that Ordnance Engineering and MS standard (commercial) parts are in accordance with the requirements of the applicable drawings.

4.3 Sampling.

4.3.1 Inspection lot.- An inspection lot shall consist of heating and tie-down units made under essentially the same conditions.

4.3.2 Parts and assemblies.- Sampling for inspection of parts and assemblies shall be as specified in the quality assurance documents listed in the Index of Supplementary Quality Assurance Provisions, LSQAP-8419330.

4.3.3 Packaging, packing and marking.- Sampling for inspection of packaging, packing and marking shall be as specified in MIL-P-14232.

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4.4 Examination.

4.4.1 Assembled units.- Each assembled heating and tie-down unit shall be examined visually for the following:

a. Completeness of assembly.
b. Presence and continuity of protective finish.
c. Presence and continuity of painting.
d. Correctness of marking.
e. Continuity of welds and conformance with weld symbols of drawing.

4.4.2 Parts and assemblies.- Examination of parts and assemblies shall be as specified in the quality assurance documents listed in the Index of Supplementary Quality Assurance Provisions, LSQAP-8419330.

4.4.3 Packaging, packing and marking.- Examination for packaging, packing and marking shall be in accordance with Specification MIL-P-14232.

4.5 Inspection testing.- Each assembled heating and tie-down unit shall be tested as follows:

4.5.1 Locking devices.- All locking devices and fastenings shall be operated manually and inspected for compliance with 3.3.1.

4.5.2 Electrical system.- Each outlet receptacle of the two receptacle box assemblies (Part Number 8419360) shall be tested, using a voltmeter, for compliance with the output requirements of 3.3.2.

4.5.3 Saddle and base assemblies.- With applicable latch assembly in unlocked position, each saddle assembly shall be functioned manually to determine compliance with 3.3.3.

4.5.4 Release assembly.- One (1) release assembly (Part Number 8417254) of each lot shall be subjected to the following test: Before securing the actuating cable to the remote control head, a testing device furnished by the contractor and approved by the Government shall be attached to the actuating cable and anchored to a convenient surface. Release the fuse link assembly from the trigger of the release assembly by depressing the trigger. The resulting force exerted on the actuating cable by the release mechanism and measured by the testing device shall conform to the requirements of 3.3.5.

4.6 Inspection equipment.

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4.6.1 Responsibility for supply, maintenance and disposition of acceptance inspection equipment prescribed on the applicable acceptance inspection equipment list shall be in accordance with Specification MIL-I-45607. Use of contractor's inspection equipment when desired by the Government inspector shall be permitted without charge. List of Government inspection equipment for examination and tests of parts for the heating and tie-down unit is contained in Index of Inspection Equipment Lists, IEL 8419330.

4.7 Rejection.- Failure of any heating and tie-down unit to pass all examinations and tests specified herein shall be cause for corrective action by the contractor.

5. PREPARATION FOR DELIVERY

5.1 Materials and methods.- All materials, methods and procedures shall be in accordance with this specification and the applicable documents. The methods of preservation for Levels A and C shall be as specified in Table I. The Government inspector shall determine that all materials meet the specification requirements. Packaging materials used in this specification are referred to as follows:

REFERENCE	IDENTIFICATION
Automotive and Artillery Grease	Specification MIL-G-10924
Corrosion Preventive Compound	Grade I of Specification MIL-C-16173
Barrier Material	Type I of Specification MIL-P-130
Grade A Barrier Material	Type I, Class 2, of Specification MIL-B-121
Osnaburg Cloth	Type I, Class 2, Finish B, of Specification CCC-C-429
Pressure-Sensitive Tape	Type I, Class I, Olive Drab of Specification PPP-T-60
Galvanized Steel Strapping	Type I, Class B of Specification QQ-S-781
Sealed Bag	Grade A, Type B-2, Class I of Specification MIL-B-13239
Sprayable Plastic Coating	Type I of Specification MIL-B-12121

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5.2 Preparation for packaging.

5.2.1 Cleaning.- All surfaces of the heating and tie-down unit shall be free of shop coatings, dirt, metal chips and other foreign material. All unpainted metal surfaces shall be cleaned in accordance with process C-1 of Specification MIL-P-116.

5.2.2 Drying.- Drying of the heating and tie-down unit shall be in accordance with D-1 drying procedure of Specification MIL-P-116.

5.2.3 Preservation.

5.2.3.1 Level A.- Preservation of the heating and tie-down unit shall be in accordance with the detailed instructions of Table I. Exposed unpainted metal surfaces of the unit not covered specifically or by inference shall receive an application of corrosion preventive compound. (See 5.1.)

5.3 Packaging.

5.3.1 Level A.- Packaging of the heating and tie-down unit shall conform to the detailed instructions of Table I. Unless otherwise specified, sprayable plastic coating shall normally be used in conjunction with Osnaburg cloth. (See 5.1.) The barrier-material shall be applied in successive layers until a thickness of 0.035 inch is obtained. The coating shall extend at least one inch onto the parent surface of the unit or to the distance specified in Table I.

5.4 Packing.

5.4.1 Level A.- Packing procedures shall conform to the detailed instructions of Table I.

TABLE I

Preservation, Packaging and Packing Procedures

Part	Packaging and Packing Required (See 5.1)
Batteries, Storage	Storage Batteries shall be shipped dry in their applicable compartment with the cable terminals disengaged from their respective battery posts. Unpainted terminals shall be coated with automotive and artillery grease, wrapped with Grade A barrier material and pressure sensitive tape utilized to secure the wraps and prevent cable movement. Battery posts shall be covered with plastic caps;

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	or pressure sensitive tape. The vent-holes in the filler caps of dry batteries shall be plugged or sealed.
Crate Blocks, Adjustable	The ratchet plate of the adjustable crate blocks shall be coated with corrosion preventive compound.
Engine, Generator and Housing Assembly	The generator engine shall be preserved in accordance with a type III engine of Specification MIL-E-10062. The AC cable assembly with the electrical cap in place shall be coiled around the shutter housing and secured with pressure sensitive tape. The lid of the housing assembly shall be closed and the latches secured. The shutter located on the end of the housing assembly, shall be covered with Osnaburg cloth, secured with pressure sensitive tape and oversprayed with sprayable plastic coating.
Fire Extinguisher	The record card shall be secured to the fire extinguisher with wire. The remote control head and valve shall be wrapped with barrier material, overwrapped with Osnaburg cloth, and the entire wrap sprayed with sprayable plastic coating.
Saddle Assemblies	The warhead and jato assemblies shall be disengaged from their respective base plates. The front and rear warhead saddle assemblies (8419390 and 8419396) and front and rear jato saddle assemblies (8419340 and 8419382) shall be packed in a class 2, style 2 nailed wood box of Specification PPP-B-621. Barrier material shall be utilized as protective insulation where container contacts components of the saddle assemblies. Closure shall be effected in accordance with applicable requirements of Specification MIL-P-12841. Store with OVE materiel.

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Strap Assemblies, Tie Down	The tie down strap assemblies shall be compactly folded and banded with pressure sensitive tape. The folded strap assemblies shall be inclosed in a sealed bag.
Disposition of Boxes	Boxes shall be secured with galvanized steel strapping inside the body, without increasing cubic displacement. Strapping used to secure containers on the vehicle shall be larger and stronger than that used to reinforce containers.

5.5 Packaging of vehicle.- The truck chassis shall be prepared for shipment in accordance with the applicable requirements of Standard MIL-STD-281.

5.6 Carloading.- Carloading and blocking procedures shall be in accordance with the Association of American Railroads Pamphlet No. MD-7, Rules Governing the Loading of Department of Defense Material on Open Top Cars.

5.7 Marking.

5.7.1 Shipping containers.- Marking of shipping containers shall be in accordance with the applicable requirements of Standard MIL-STD-129.

5.7.2 Vehicle markings.- Marking requirements for the vehicle shall be as specified in Standard MIL-STD-281.

5.8 Repair parts.- Preservation, packaging, packing and marking of repair parts shall be in accordance with the applicable packaging data sheets of Specification MIL-W-14539.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Title, number and date of applicable drawings.
c. List of Government furnished equipment.
d. Place of inspection.
e. Place of test.
f. List of maintenance and repair parts required with title, number and date of applicable Supplementary Quality Assurance Provisions.
g. Any exceptions to the applicable Supplementary Quality Assurance Provisions.

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6.2 Suggested contractual features.

6.2.1 Serial numbers.- The contractor should be furnished a block of serial numbers to be used.

6.2.2 Color.- Chip No. 24087 of Federal Standard 595, Colors, should be furnished the contractor to be used as a color sample for painting (see 3.4).

Notice.- When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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